Atomics and the Memory Model

Atomic operations, compare-and-swap, memory orders and happens-before with a simulator, the ordering patterns that matter, and thread-safe data structures from fine-grained locks to lock-free. Builds on Multithreading.

Atomic operations and `compare_exchange`

`std::atomic<T>` makes operations on **one variable** indivisible, usually without blocking. The increment is now one step:

SIMULATOR **std::atomic: each increment is a single indivisible step**

A `r = 0`

```
counter.fetch_add(1)
counter.fetch_add(1)
```

Step A

B `r = 0`

```
counter.fetch_add(1)
counter.fetch_add(1)
```

Step B

shared counter
0

There is no gap between the read and the write for another thread to sneak into.

Reset | Finish randomly | Run 500 random schedules

The scheduler may switch threads between any two instructions.

```cpp
#include <atomic>

std::atomic<int> counter{0};
counter++;                                  // one indivisible read-modify-
write
counter.fetch_add(1);                       // same thing, explicit
int v = counter.load();                     // atomic read
counter.store(5);                           // atomic write
int old = counter.exchange(9);              // write 9, get the previous
value, all in one step

std::atomic<bool> stop{false};              // a very common use: a shutdown
flag
```

USE AN ATOMIC FOR	DO **NOT** USE IT FOR
a counter, flag, swapped pointer, statistics	an invariant over **several** variables: use a mutex

Compare-and-swap (CAS)

`compare_exchange(expected, desired)`: "if the value still equals `expected`, set it to `desired`; otherwise load the current value into `expected` and fail". It is the building block of every lock-free update.

```cpp
int expected = x.load();
while (!x.compare_exchange_weak(expected, expected * 2)) {
    // someone changed x: `expected` now holds the current value, so
recompute and retry
}
```

	`COMPARE_EXCHANGE_WEAK`	`COMPARE_EXCHANGE_STRONG`
spurious failure	**may** fail even when the value matched	fails only if the value really differed
use	inside a retry **loop** (cheaper on some CPUs)	a single attempt, no loop

Compiled: a CAS with a stale `expected` returned false and refreshed `expected` to the current value (99).

`atomic_flag` and a spinlock

```
struct Spinlock {
    std::atomic_flag f = ATOMIC_FLAG_INIT;
    void lock()   { while (f.test_and_set(std::memory_order_acquire)) { /*
spin */ } }
    void unlock() { f.clear(std::memory_order_release); }
};
```

`atomic_flag` is the one type guaranteed lock-free. Four threads each doing 50,000 increments under this spinlock reached exactly 200,000 (compiled). A spinlock burns a core while waiting: fine for tiny critical sections on dedicated cores, bad when the lock holder can be descheduled (use a mutex).

> **VOLATILE IS NOT THREAD-SAFE** It only stops the compiler optimising an access away. No atomicity, no ordering. Use `std::atomic`.

- `is_lock_free()` says whether the type is done with real atomic instructions; big structs fall back to a hidden lock.
- **ABA problem:** a CAS can succeed when the value went A to B and back to A, although the world changed meanwhile. Fix with a version counter next to the value, or hazard pointers.

> **READ MORE**
>
> **Effective Modern C++ (Meyers)**
>
> - Item 40, Use std::atomic for concurrency, volatile for special memory (PDF pages 289 to 297)
>
> **C++ Concurrency in Action (Williams)**
>
> - 5.2 Atomic operations and types in C++ (atomic_flag, atomic<bool>, pointers, integers, free functions) (PDF pages 151 to 164)
> - 7.1 to 7.3 Lock-free data structures: definitions, a lock-free stack, memory reclamation, the ABA problem (PDF pages 229 to 274)

Memory orders and happens-before

Compilers and CPUs **reorder** memory operations, and each core has buffers and caches, so one thread's write may become visible to another thread *later*, or *after* a write that came after it. A plain `int` gives no guarantee. Atomics let you say how much ordering you need.

The one rule to learn: release and acquire

```cpp
int data = 0;
std::atomic<bool> ready{false};

// producer thread
data = 42;
ready.store(true, std::memory_order_release);

// consumer thread
while (!ready.load(std::memory_order_acquire)) { }
std::cout << data;                          // guaranteed to print 42
```

- A **release store** publishes: every write the thread made *before* it becomes visible to any thread that reads the stored value with an **acquire load**.
- The release/acquire pair **synchronizes-with**, and that creates a **happens-before** edge: `data = 42` happens before the consumer's read, so the read is not a data race and sees 42.
- **happens-before** is the formal "must be visible" relation. Within one thread it is program order; across threads it comes only from synchronization (atomics, mutexes, `join`, futures).

Try it. Flush the producer's writes to memory in different orders and see what the consumer prints:

SIMULATOR **Publishing data with a flag: what can the consumer see?**

store relaxed load relaxed without both release and acquire, C may print 0

P: producer

```
data = 42;
ready.store(true, relaxed);
```

Step P

C: consumer

```
while (!ready.load(relaxed)) {}
print(data);
```

Step C

Producer store buffer (not yet visible):

(empty)

Shared memory:

shared data	shared ready
0	false

Step P to queue its writes, flush them to memory, and step C to read. Try the two orders above.

Reset | Run 500 random schedules

Simplified model: it shows what the C++ memory model allows. x86 hardware often hides these outcomes.

The orders

ORDER	GUARANTEE	USE
`relaxed`	only atomicity (and a single total order for *that* variable)	counters and statistics nobody synchronizes on
`acquire` (loads)	nothing after it moves before it; sees what the matching release published	reading a flag or pointer that publishes data
`release` (stores)	nothing before it moves after it; publishes earlier writes	setting that flag or pointer

`acq_rel` (read-modify-write)	both at once	`fetch_sub` on a reference count, a lock hand-over
`seq_cst` (default)	acquire/release **plus** one global order of all `seq_cst` operations that every thread agrees on	when in doubt: it is the only order that is easy to reason about

X86 HIDES BUGS x86 already gives release/acquire for ordinary loads and stores, so wrong orderings often "work" on a developer's PC and fail on ARM (phones, Apple silicon, many servers). The simulator above shows what the **C++ model allows**, not what x86 does.

- Both sides must take part: a release store read by a **relaxed** load publishes nothing.
- A **fence** (`std::atomic_thread_fence`) gives the ordering without tying it to one variable. Know it exists; reach for acquire/release on the atomic first.
- Without any synchronization, two threads touching the same non-atomic variable (one writing) is a **data race: undefined behaviour**, not merely "stale data".

KEY POINT "What does acquire/release guarantee?" Everything written before the release is visible after the matching acquire. Say *synchronizes-with* and *happens-before*, and that `seq_cst` is the safe default.

READ MORE

C++ Concurrency in Action (Williams)

- 5.1 Memory model basics (objects, memory locations, modification orders) (PDF pages 148 to 150)
- 5.3 Synchronizing operations and enforcing ordering (synchronizes-with, happens-before, the memory orders, release sequences, fences) (PDF pages 165 to 196)

Ordering patterns that matter

PATTERN	ORDERS	WHY

Publish with a flag or pointer	store `release`, load `acquire`	the data written before the store is visible after the load
Statistics counter	`relaxed` `fetch_add`	only the final total matters; no other data depends on it
Reference count	increment `relaxed`, decrement `acq_rel`	adding a reference needs no ordering; the thread that drops the last one must see every other thread's writes before it deletes
Spinlock	lock `acquire`, unlock `release`	a critical section must not leak out of the lock in either direction
Shutdown flag	`relaxed` if nothing else depends on it, else `release` / `acquire`	a flag alone only needs to be seen eventually
SPSC queue indices	own index `relaxed`, other's `acquire`, publish `release`	see the ring buffer in Performance & Memory

```
struct Counted {                                                    // like
shared_ptr's control block
    std::atomic<int> refs{1};
    void addRef()  { refs.fetch_add(1, std::memory_order_relaxed); }
    void release() {
        if (refs.fetch_sub(1, std::memory_order_acq_rel) == 1)    // the last
one out sees everyone's writes …
            delete this;                                            //  … before
destroying the object
    }
};
```

Compiled: five owners on four threads, the object was destroyed exactly once; a 2,000-round release/acquire publish test read a wrong value 0 times.

Double-checked locking (the classic trap)

```
std::atomic<Widget*> inst{nullptr};
std::mutex m;
Widget* get() {
    Widget* p = inst.load(std::memory_order_acquire);            // 1st
check, no lock
    if (!p) {
        std::lock_guard lk(m);
        p = inst.load(std::memory_order_relaxed);                // 2nd
check, under the lock
        if (!p) { p = new Widget; inst.store(p, std::memory_order_release); }
// publish a FULLY built object
    }
    return p;
}
```

Without the release/acquire pair, another thread could see the pointer before the constructor's writes. In practice use a function-local `static` or `std::call_once` and let the compiler do this.

> **BE HONEST ABOUT WHAT YOU KNOW** Start with a mutex or `seq_cst`, measure, and weaken orders only when you can say which happens-before edge each one provides. "I use acquire/release for publication, relaxed for counters" is a sound rule of thumb.

> **READ MORE**
>
> **C++ Concurrency in Action (Williams)**
>
> - 5.3 Synchronizing operations and enforcing ordering (synchronizes-with, happens-before, the memory orders, release sequences, fences) (PDF pages 165 to 196)

Thread-safe data structures

Three levels, in the order you should try them:

LEVEL	HOW	EXAMPLE

1. One coarse lock	a mutex around the whole structure	a stack or map with a mutex (see the interface races in Multithreading)
2. Fine-grained locks	different mutexes for different parts, so threads that touch different parts do not wait	a queue with a head mutex and a tail mutex; a hash table with one mutex per bucket
3. Lock-free	atomics and CAS loops, no mutex	a Treiber stack, an SPSC ring buffer

Fine-grained: a queue with two locks

```
template <typename T> class TwoLockQueue {
    struct N { std::shared_ptr<T> data; std::unique_ptr<N> next; };
    std::mutex headM_, tailM_;                      // producers take tailM_,
consumers take headM_
    std::unique_ptr<N> head_;
    N* tail_;
public:
    TwoLockQueue() : head_(new N), tail_(head_.get()) {}   // a DUMMY node:
head and tail never meet on the same node
    void push(T v);                                         // locks tailM_
only
    std::optional<T> tryPop();                              // locks headM_
only (and tailM_ briefly to read the tail)
};
```

The **dummy node** is the trick: with it, a push only touches the tail and a pop only the head, so one producer and one consumer never block each other. Compiled: a producer pushing 1 to 1000 and a consumer popping them summed to 500500.

Read-mostly: a lookup table

A price table read by many threads and updated rarely: `std::shared_mutex` with `shared_lock` for readers and `unique_lock` for writers (Other mutex types). Return **copies** or small values, never references into the table.

Lock-free: a stack push

```
struct Node { int value; Node* next; };
std::atomic<Node*> head{nullptr};

void push(int v) {
    Node* n = new Node{v, head.load(std::memory_order_relaxed)};
    while (!head.compare_exchange_weak(n->next, n, std::memory_order_release,
std::memory_order_relaxed)) {}
    // on failure compare_exchange_weak reloads n->next with the current
head: just retry
}
```

Compiled: four threads pushing 1000 nodes each gave 4000 nodes, no lock. **Pop is the hard part**: another thread may pop and delete the node you are reading (**memory reclamation**), and the ABA problem can corrupt the list. Real implementations need hazard pointers, epochs or a counted pointer.

> LOCK-FREE IS NOT FASTER BY DEFAULT A contended CAS loop can be slower than a mutex, and a bug shows up once a month on a customer machine. Use a mutex first, measure, and use a proven library (or the SPSC ring when one producer and one consumer is enough) before writing your own.

- **Wait-free** (every thread finishes in a bounded number of steps) is stronger than **lock-free** (some thread always makes progress) which is stronger than **obstruction-free**.
- Design the **interface** first: whole operations (`tryPop` returning an `optional`), not `empty()` then `pop()`.

> READ MORE
>
> **C++ Concurrency in Action (Williams)**
>
> - 6.2 Lock-based concurrent data structures (a thread-safe stack and queue, fine-grained locks) (PDF pages 199 to 216)
> - 7.1 to 7.3 Lock-free data structures: definitions, a lock-free stack, memory reclamation, the ABA problem (PDF pages 229 to 274)

Cheat sheet

Say each answer aloud, then open it.

<details>
<summary>▼ <code>std::atomic</code> vs mutex?</summary>

An atomic makes operations on a single variable indivisible, usually lock-free and cheap. A mutex protects any critical section, including invariants over several variables, but may block. Use atomics for counters and flags, mutexes for compound state.
</details>

<details>
<summary>▼ Is <code>volatile</code> a substitute for <code>atomic</code>?</summary>

No. <code>volatile</code> prevents some compiler optimisations but gives no atomicity and no inter-thread ordering. Using it for synchronization is a data race.
</details>

<details>
<summary>▼ What are the memory orders?</summary>

<code>seq_cst</code> (default, strongest, one global order), <code>acquire</code> / <code>release</code> (pair a release store with an acquire load to publish data), <code>acq_rel</code> for read-modify-write, and <code>relaxed</code> (atomicity only). Start with the default; weaken only with a reason and a clear happens-before argument.
</details>

<details>
<summary>▼ What does release/acquire guarantee?</summary>

Everything the producer wrote before a release store is visible to a thread that reads the stored value with an acquire load. The pair synchronizes-with, which gives a happens-before edge, so the earlier plain writes are not a data race.
</details>

▼ What does happens-before mean?

If A happens-before B, then A's effects are visible to B. Within a thread it is program order; between threads it exists only through synchronization: atomics with matching orders, mutex unlock/lock, thread start and join, futures. Two conflicting accesses without a happens-before relation are a data race.

▼ When is `relaxed` enough?

When no other data depends on the atomic: a statistics counter, an id generator, a shutdown flag that carries no data. The final value is correct, but it orders nothing else.

▼ Why can code with a wrong memory order work on x86 and fail on ARM?

x86's hardware already orders ordinary loads and stores like release/acquire, so missing orderings are hidden. ARM and POWER reorder far more, so the same code can read stale data there. The C++ model, not the PC you tested on, defines correctness.

▼ `compare_exchange_weak` vs `strong`?

Both set a value only if it still equals the expected one. `weak` may fail spuriously, so it belongs in a retry loop and is cheaper on some CPUs; `strong` fails only on a real mismatch and suits a single attempt. On failure they load the current value into `expected`.

▼ What is the ABA problem?

A CAS succeeds because the value is A again, but it went A to B to A in between, and the structure around it changed (for example a node was freed and reused). Fix with a version counter beside the pointer, or hazard pointers or epoch reclamation.

▼ When is a spinlock appropriate?

For very short critical sections on dedicated cores where waiting is shorter than a context switch. It burns CPU while waiting and is dangerous if the holder can be descheduled. Otherwise use a mutex.

▼ How is a reference count made thread-safe, and with which orders?

With an atomic counter: increments can be relaxed, the decrement is `acq_rel` so the thread that drops it to zero sees all other threads' writes before it destroys the object. `shared_ptr`'s control block works this way.

▼ Mutex, fine-grained locks or lock-free: how do you choose?

Start with one mutex and measure. Split locks if contention shows up and the structure has independent parts (head and tail of a queue, buckets of a table). Go lock-free only for a proven bottleneck and with a known-correct design, because memory reclamation and ABA make it hard; an SPSC ring buffer is the one case simple enough to write confidently.